NAME OF CANDIDATE: .....
STUDENT NUMBER: .....
COURSE (3871 or 5960): .....
TUTORIAL GROUP (day & time): .....

THE UNIVERSITY OF NEW SOUTH WALES

SCHOOL OF MATHEMATICS AND STATISTICS

Mid-Session Test, T3 2025

MATH3871/5960

Bayesian Inference and Computation

(1) TIME ALLOWED – 1.5 hours + 10 mins reading time

(2) TOTAL NUMBER OF QUESTIONS – 4

(3) ANSWER ALL QUESTIONS

(4) THE QUESTIONS ARE OF EQUAL VALUE

(5) WRITE YOUR ANSWERS IN THE SPACE PROVIDED

(6) THIS PAPER MAY **NOT** BE RETAINED BY THE CANDIDATE

(7) ONLY CALCULATORS WITH AN AFFIXED "UNSW APPROVED" STICKER MAY BE USED.

All answers must be written in ink. Except where they are expressly required pencils may only be used for drawing, sketching or graphical work.

1. **[10 marks]** Let x be the number of successes in n independent trials with probability p of success in each trial. Suppose that both the probability of success p is unknown and the number of trials n is unknown (i.e. the number of trials that can be conducted depends on some random process). The likelihood function is $L(x|n,p) = \binom{n}{x}p^x(1-p)^{n-x}$. Let n and p have independent prior distributions, i.e. $\pi(n,p) = \pi(n)\pi(p)$. Suppose both $\pi(n)$ and $\pi(p)$ are uniform priors (i.e. $\pi(n) \propto 1$ and $\pi(p) \propto 1$).

Hint: Recall that for positive integers n, x,

$$\binom{n}{x} = \frac{n!}{x!(n-x)!} \qquad \textit{and} \qquad \Gamma(n) = (n-1)!$$

a) **[1 marks]** Are $\pi(n)$ and $\pi(p)$ proper or improper priors? Explain why.

b) **[3 marks]** Give the joint posterior distribution $\pi(n,p|x)$ and the marginal posterior distribution $\pi(n|x)$. You may provide your answers in unnormalised form.

c) **[2 marks]** Based on your answer in (b), explain whether the chosen prior $\pi(n)$ is acceptable to use or not.

d) **[2 marks]** Suppose that

$$\hat{\pi}(\theta|x) = \frac{1}{N}\sum_{i=1}^{N}\delta_{\theta^{(i)}}(\theta)$$

is an approximation of the posterior distribution, where $\theta^{(1)}, \ldots, \theta^{(N)} \sim \pi(\theta|x)$, and $\delta_b(A) = 1$ if $b \in A$ and 0 otherwise. Show that an approximation to an expectation with respect to the posterior is given by

$$E_\pi[g(\theta)] \approx \frac{1}{N}\sum_{i=1}^{N} g(\theta^{(i)}).$$

e) **[2 marks]** Suppose that $\theta^{(1)}, \ldots, \theta^{(N)} \sim \pi(\theta|x)$, and that $\bar{g} = \frac{1}{N}\sum_{i=1}^{N} g(\theta^{(i)})$ and $s_g^2 = \frac{1}{N}\sum_{i=1}^{N}(g(\theta^{(i)}) - \bar{g})^2$. How many samples, $g(\theta^{(i)})$, are required to ensure that the Monte Carlo estimate, $\bar{g}$, is within 0.01 of its true value with probability $1-\alpha$, for $\alpha \in (0,1)$?

Solutions

a) **[1 mark]** $\pi(p)$ is proper but $\pi(n)$ is improper. That's because the support of $\pi(p)$ is compact so a uniform prior will integrate to a constant, whereas the support of $\pi(n)$ is unbounded, so it will not sum to a constant.

Please see over ...

b) **[3 marks]** The joint posterior distribution is

$$\pi(n,p|x) \propto \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} \times 1 \times 1 & n \geq x \\ 0 & \text{otherwise} \end{cases}$$

(1 mark for correctly identifying the two cases, since we can't have more observations than trials.)
For the posterior marginal: using the above

$$\pi(n,p|x) \propto \binom{n}{x} p^x (1-p)^{n-x} \times 1 \times 1$$

for $n \geq x$, then integrate wrt p which is a cts random variable,

$$\begin{aligned} \pi(n|x) \propto & \binom{n}{x} \int_0^1 p^x (1-p)^{n-x} dp \\ &= \binom{n}{x} \frac{\Gamma(x+1)\Gamma(n-x+1)}{\Gamma(n+2)} \\ &= \frac{n!}{x!(n-x)!} \frac{x!(n-x)!}{(n+1)!} \\ &= \frac{1}{n+1} \end{aligned}$$

where we recall the second line comes from noticing that the integrand is proportional to a beta density.

c) **[2 marks]** The prior $\pi(n)$ is not admissible because the marginal posterior $\pi(n|x)$ is clearly improper (1 mark) since $\sum_n \pi(n|x) = \sum_n \frac{1}{n+1} \to \infty$ (1 mark).

d) **[2 marks]** We have

$$\begin{aligned} E_\pi[g(\theta)] &= \int g(\theta)\pi(\theta|x)d\theta \\ &\approx \int g(\theta) \frac{1}{N} \sum_{i=1}^N \delta_{\theta^{(i)}}(\theta) d\theta \\ &= \frac{1}{N} \sum_{i=1}^N g(\theta^{(i)}). \end{aligned}$$

e) **[2 marks]** We want one half of a $(1-\alpha)\%$ confidence interval to be less than 0.01.
Hence $\frac{s_g}{\sqrt{N}} t_{N-1,1-\alpha/2} \leq 0.01$. Approximating the t by a standard normal, and solving for N, we get

$$N \geq \left(\frac{s_g z_{1-\alpha/2}}{0.01} \right)^2.$$

Please see over ...

2. [10 marks]

a) **[3 marks]** Determine the form of a conjugate prior for the exponential family of models, given by

$$f(x|\theta) = h(x)g(\theta)\exp\{t(x)c(\theta)\},$$

for functions h, g, t, and c such that $\int f(x|\theta) = 1$.

b) **[2 marks]** Hence, using the result from part a), *derive* the form of the conjugate prior when $X \sim \text{Binomial}(n, \theta)$.

c) **[1 marks]** When is it appropriate and when is it not appropriate to use conjugate priors?

d) **[1 marks]** Do conjugate priors exist for a model $h(x|\theta)$ in *any* family? Explain why. If so, are there any conditions?

e) **[2 marks]** Suppose that X is a random variable with distribution function $F(x)$. By calculating the cdf, show that the distribution of $U = F(X)$ is U(0,1). Hence, deduce a method of generating samples from $F(x)$ given draws $u^{(1)}, \ldots, u^{(N)} \sim U(0, 1)$.

f) **[1 marks]** Give one benefit and one weakness of inversion sampling.

Solutions

a) **[3 marks]** With prior $\pi(\theta)$ we have

$$\begin{aligned}\pi(\theta|x) &\propto L(x|\theta)\pi(\theta)\\ &= \pi(\theta)\prod_{i=1}^{n} h(x_i)g(\theta)\exp\{t(x_i)c(\theta)\}\\ &\propto \pi(\theta)g(\theta)^n\exp\{c(\theta)\sum_i t(x_i)\}.\end{aligned}$$

This means if we choose $\pi(\theta) \propto g(\theta)^d \exp\{bc(\theta)\}$ for some d, b we would get

$$\begin{aligned}\pi(\theta|x) &\propto g(\theta)^{d+n}\exp\{c(\theta)[\sum_i t(x_i) + b]\}\\ &= g(\theta)^{\tilde{d}}\exp\{\tilde{b}c(\theta)\},\end{aligned}$$

which is in the exponential family (with modified parameters).

b) **[2 marks]** Here we have

$$\begin{aligned}f(x|\theta) &= \binom{n}{x}\theta^x(1-\theta)^{n-x}\\ &= \binom{n}{x}(1-\theta)^n\exp\left\{x\log\left(\frac{\theta}{1-\theta}\right)\right\}.\end{aligned}$$

Please see over ...

So we have $h(x) = \binom{n}{x}$, $g(\theta) = (1-\theta)^n$, $t(x) = x$ and $c(\theta) = \log\left(\frac{\theta}{1-\theta}\right)$.
Hence we need a prior of the form

$$\begin{aligned}\pi(\theta) &\propto [(1-\theta)^n]^d \exp\left\{b \log\left(\frac{\theta}{1-\theta}\right)\right\} \\ &= (1-\theta)^{nd-b}\theta^b\end{aligned}$$

which is in the Beta family.

c) **[1 mark]** Appropriate to use when we would like a closed-form solution, and when the conjugate prior coincides with your prior beliefs. Not appropriate when the conjugate prior is unable to express your prior beliefs.

d) **[1 mark]** In principle, a conjugate prior exists for any model in that one just chooses a prior form that follows the algebraic form of θ in the likelihood. The conditions are that the prior is defined over the full range of θ values, and that the posterior is proper.

e) **[2 marks]** Clearly we have $u \in [0,1]$ as F is a cdf. The cdf of U is then

$$\begin{aligned}Pr(U \leq u) &= Pr(F(X) \leq u) \\ &= Pr(X \leq F^{-1}(u)) \\ &= F(F^{-1}(u)) = u\end{aligned}$$

which is the cdf of a $U(0,1)$ random variable.
This means an algorithm to sample from $F(x)$ could be: set $x^{(i)} = F^{-1}(u^{(i)})$ where $u^{(i)} \sim U(0,1)$.

f) **[1 mark]** Benefit = e.g. independent samples. Weakness = e.g. restricted to univariate distributions.

3. [10 marks]

a) **[3 marks]** By using a suitable transformation that maps the integration limits to $[0,1]$, write the integral $\int_a^\infty g(x)dx$, for $0 \leq a \leq \infty$, in the form $\int_0^1 g'(u)du$. Hence write down a Monte Carlo estimate of this integral using samples $u^{(1)}, \ldots, u^{(N)} \sim U(0,1)$.

b) **[2 marks]** The rejection sampling algorithm can be written in two ways (given below). By considering the overall acceptance probability of each algorithm, explain why these two versions are equivalent.

Rejection sampling (version 1)

i) Generate x^* from $g(x)$
ii) Generate y^* from $U(0, Kg(x^*))$
iii) Accept x^* if $y^* \leq f(x^*)$

Please see over ...

Rejection sampling (version 2)

i) Generate x^* from $g(x)$

ii) Accept x^* with probability $\frac{f(x^*)}{Kg(x^*)}$

where in each case, $f(x) \leq Kg(x)$, $\forall x$.

c) **[3 marks]** Let $X \sim g(x)$ such that $f(x) \leq Kg(x)$, $\forall x$, for some K. By evaluating $\Pr(X \leq x | X \text{ is accepted})$, show that samples accepted under rejection sampling follow the distribution $f(x)$. Assume that $\int f(y)dy = c < \infty$.

d) **[2 marks]** Hence, or otherwise, calculate the overall acceptance rate of the algorithm. How does the overall acceptance rate inform the optimal choice of K?

Solutions

a) **[3 marks]** A number of transformations can work here, like $u = a/x$ and similar. Another option is to use two transformations, one to map $[a, \infty) \to [0, \infty)$, such as $u = x - a$, and then another to map $[0, \infty) \to [0, 1]$, like $v = 1/(u + 1)$. Here we use the substitution $u = 1/(x - a + 1)$. We then have $x = 1/u + a - 1$, and $dx = -1/u^2 du$. Hence we then have

$$\int_a^\infty g(x)dx = \int_0^1 g(1/u + a - 1)/u^2 du.$$

Hence a Monte Carlo estimate of the integral is

$$\int_a^\infty g(x)dx \approx \frac{1}{N} \sum_i g(1/u^{(i)} + a - 1)/[u^{(i)}]^2 \quad \text{where } u^{(1)}, \ldots, u^{N)} \sim U(0, 1).$$

b) **[2 marks]** Both versions draw $x^* \sim g(x)$. We then have

$$\begin{aligned} Pr(\text{acceptance under version 2}) &= \frac{f(x^*)}{Kg(x^*)} \quad \text{(directly)} \\ Pr(\text{acceptance under version 1}) &= Pr(y^* \leq f(x^*)) \\ &= \frac{f(x^*)}{Kg(x^*)} - \frac{0}{Kg(x^*)} = \frac{f(x^*)}{Kg(x^*)}, \end{aligned}$$

because $y^* \sim U(0, Kg(x^*))$ for which $Pr(y^* < y) = \frac{y}{Kg(x^*)}$. Hence both algorithm versions have the same acceptance probability.

c) **[3 marks]** Let $h(x) = f(x)/Kg(x)$ be the probability that x is accepted, and note that $\int f(x)dx = c$ is not a normalised cdf. Then

$$\Pr(X \leq x \text{ and } X \text{ accepted}) = \int_{-\infty}^x h(y)g(y)dy$$

Please see over ...

and

$$\Pr(X \text{ accepted}) = \int_{-\infty}^{\infty} h(y)g(y)dy$$

so

$$\begin{aligned}\Pr(X \le x | X \text{ accepted}) &= \frac{\int_{-\infty}^{x} h(y)g(y)dy}{\int_{-\infty}^{\infty} h(y)g(y)dy} \\ &= \frac{\int_{-\infty}^{x} f(y)dy}{\int_{-\infty}^{\infty} f(y)dy} = F(x),\end{aligned}$$

which is the cdf of $f(x)$. That is, the accepted values have pdf $f(x)$.

d) The overall acceptance rate is:

$$\Pr(X \text{ accepted}) = \int h(y)g(y)dy = \frac{1}{K}\int f(y)dy \propto \frac{1}{K}.$$

Hence want K as small as possible to increase the overall acceptance rate. But this must be subject to $f(x) \le Kg(x)$. I.e. $K = \max_x f(x)/g(x)$

4. [10 marks]

a) **[3 marks]** Show that the expected posterior loss,

$$E_\pi[L(\theta, d)] = \int L(\theta, d)\pi(\theta|x)d\theta$$

is minimised at the posterior mean $d = E(\theta|x)$ when $L(\theta, d) = (\theta - d)^2$ is the quadratic loss function.

b) **[2 marks]** In a particular Bayesian analysis, where the loss function quantifies the \$ value in costs of some industrial process, the posterior variance is estimated as $\hat{V} = 2.8$ million. Under quadratic loss, if a company is willing to tolerate expected costs of \$C or less, what range of point estimates, d, can they use?

c) **[1 marks]** Can the company achieve their goal if \$C = \$2.3 million?

d) **[3 marks]** Suppose that $x_1, \ldots, x_n$ follow the Pareto distribution, Pareto(a, b), with known a, which has density function

$$f(x|b) = ba^b x^{-b-1} \qquad \text{for } x > a.$$

Find the asymptotic posterior distribution for b.

e) **[1 marks]** How would the asymptotic posterior in part d) change for different prior distributions for b?

Please see over ...

Solutions:

a) **[3 marks]**

$$\begin{aligned}
E_\pi[L(\theta,d)] &= \int L(\theta,d)\pi(\theta|x)d\theta \\
&= \int (\theta-d)^2\pi(\theta|x)d\theta \\
&= \int (\theta-E(\theta|x)+E(\theta|x)-d)^2\pi(\theta|x)d\theta \\
&= \int (\theta-E(\theta|x))^2\pi(\theta|x)d\theta + \int (E(\theta|x)-d)^2\pi(\theta|x)d\theta \\
&\quad +2\int (\theta-E(\theta|x))(E(\theta|x)-d)\pi(\theta|x)d\theta \\
&= Var(\theta|x) + (E(\theta|x)-d)^2 + 0.
\end{aligned}$$

This is clearly minimised when $d = E(\theta|x)$.

b) **[2 marks]** To achieve $E_\pi[L(\theta,d)] < C$ we need:

$$\begin{aligned}
E_\pi[L(\theta,d)] &\leq C \\
V + (E(\theta|x)-d)^2 &\leq C \\
(E(\theta|x)-d)^2 &\leq C-V \\
-\sqrt{C-V} \leq E(\theta|x)-d &\leq \sqrt{C-V} \qquad \text{note: } C > V \text{ is required} \\
E(\theta|x)-\sqrt{C-V} \leq d &\leq E(\theta|x)+\sqrt{C-V}.
\end{aligned}$$

c) **[1 mark]** The minimum value of the expected loss is $\text{Var}(\theta|x) = 2.8$m. Hence we can't have an expected loss lower than this, and the company is unable to achieve their $\$C <\$2.8$m cost goal.

d) **[3 marks]**

$$\begin{aligned}
\log L(x|\theta) &= n\log b + bn\log a + (-b-1)\sum_i \log x_i \\
\partial \log L(x|\theta)/\partial b &= \frac{n}{b} + n\log a - \sum_i \log x_i \\
\partial^2 \log L(x|\theta)/\partial b^2 &= -\frac{n}{b^2} \\
\hat{b} &= \frac{n}{\sum_i \log x_i - n\log a} \\
I_n(b) &= \frac{n}{b^2} \\
I_n(\hat{b}) &= \frac{(\sum_i \log x_i - n\log a)^2}{n}.
\end{aligned}$$

Hence the asymptotic distribution of b is

$$N\left(\frac{n}{\sum_i \log x_i - n\log a}, \frac{n}{(\sum_i \log x_i - n\log a)^2}\right).$$

Please see over ...

e) The asymptotic posterior is unaffected by specific choices of prior distribution, unless these give zero probability to parts of the parameter space.

Some useful definitions and results

- If $X \sim \text{Beta}(\alpha, \beta)$, $\alpha > 0, \beta > 0$, $x \in (0,1)$, $p(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}$. $E(X) = \alpha/(\alpha+\beta)$, $Var(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$.
- If $X \sim \text{Gamma}(\alpha, \beta)$, $\alpha, \beta > 0, x \geq 0$, $p(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$. $E(x) = \alpha/\beta$, $Var(X) = \alpha/\beta^2$.
- For positive integers α, $\Gamma(\alpha) = (\alpha-1)!$
- If $X \sim \text{Binomial}(n, p)$, $p \in [0,1]$, $P(X = k) = \left(\begin{array}{c} n \\ k \end{array} \right) p^k (1-p)^{n-k}$, for $k = 0, \ldots, n$. $E(X) = np$, $Var(X) = np(1-p)$.
- If $X \sim \text{Poisson}(\lambda), \lambda > 0$, $P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$ for $k = 0, \ldots,$. $E(X) = \lambda$, $Var(X) = \lambda$.
- If $X \sim N(\theta, \sigma^2)$, $p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{1}{2\sigma^2}(x-\theta)^2)$, $E(X) = \theta, Var(X) = \sigma^2$.
- If $X \sim \text{Uniform}(a, b), b > a$, $p(x) = 1/(b-a)$, $E(X) = \frac{1}{2}(a+b)$, $Var(X) = \frac{1}{12}(b-a)^2$.
- If $X \sim Exp(\lambda)$ with $\lambda > 0$, $p(x) = \lambda \exp(-\lambda x), \quad x \geq 0$, $E(X) = \frac{1}{\lambda}$ and $Var(X) = \frac{1}{\lambda^2}$.

Some standard conjugate prior-likelihood pairs.

Likelihood	Prior	Posterior
$x \sim Binomial(n, \theta)$	$Beta(p, q)$	$Beta(p + x, q + n - x)$
$x_1, \ldots, x_n \sim Poisson(\theta)$	$Gamma(p, q)$	$Gamma(p + \sum_{i=1}^n x_i, q + n)$
$x_1, \ldots, x_n \sim N(\theta, \tau^{-1})$, ($\tau$ known)	$N(b, c^{-1})$	$N(\frac{cb+n\tau\bar{x}}{c+n\tau}, \frac{1}{c+n\tau})$
$x_1, \ldots, x_n \sim Gamma(k, \theta)$ (k known)	$Gamma(p, q)$,	$Ga(p + nk, q + \sum_{i=1}^n x_i)$
$x_1, \ldots, x_n \sim Geometric(\theta)$	$Beta(p, q)$	$Beta(p + n, q + \sum_{i=1}^n x_i - n)$
$x \sim NegBinomial(r, \theta)$	$Beta(p, q)$	$Beta(p + r, q + x)$